I-Da Chiang (江宜達)

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SUMMARY

Experienced astrophysics researcher transitioning to data/ML roles. Strong background in Python-based data processing, statistical modeling, and pipeline design. Skilled in cross-disciplinary , cross-culture collaboration and technical communication. Exploring opportunities in machine learning and software development.

SKILLS

- **Programming:** Python (NumPy, pandas, scikit-learn, Keras), Git, Bash, LaTeX
- **Data Science:** data preprocessing, modeling, pipeline automation, visualization (Matplotlib, seaborn)
- **AI Tools:** Generative AI (ChatGPT, Grok, NotebookLM)
- **Collaborative Tools:** Google Workspace, Overleaf, Slack, Zoom
- **Soft Skills:** scientific writing, collaborative research, project presentation

EXPERIENCE

National Taiwan Normal University
- Full-time Assistant *2025 -*
  Organizing the 47th International School for Young Astronomers, an IAU-led initiative.

Institute of Astronomy and Astrophysics, Academia Sinica
- Postdoc Fellow *2021 - 2025*
  Supervisor: Dr. Hiroyuki Hirashita & Dr. Chian-Chou Chen
  Research topics: dust evolution, tracing molecular gas, and HI 21 cm observations

EDUCATION

University of California San Diego
- Ph.D. (Physics) *2014 - 2021*
  Thesis: "Observations of Spatially Resolved Dust Evolution in Nearby Galaxies"
  Adviser: Prof. Karin M. Sandstrom

National Taiwan University
- M.S. (Physics) *2012 - 2014*
  Thesis: "Plasmonic Enhanced Optical Disk Reactor for Wastewater Treatment"
  Adviser: Prof. Din Ping Tsai
- B.S. (Physics) *2007 - 2011*

ACADEMIC PUBLICATIONS

I published 9 first- or second-author journal papers; 29 journal papers in total. h-index: 19.

9) **I-Da Chiang** et al., "CO-to-H_2 conversion factor and grain size distribution through the analysis of $\alpha_{\rm CO}$-$q_{\rm PAH}$ relation", 2025, MNRAS, 536, 2392.

8) **I-Da Chiang** et al., "Resolved Measurements of the CO-to-H_2 Conversion Factor in 37 Nearby Galaxies", 2024, ApJ, 964, 18.

7) Yu-Hsuan Teng, **I-Da Chiang** et al., "Star Formation Efficiency in Nearby Galaxies Revealed with a New CO-to-H2 Conversion Factor Prescription", 2024, ApJ, 961, 42.

6) **I-Da Chiang** et al., "Kpc-scale properties of dust temperature in terms of dust mass and star formation activity", 2023, MNRAS, 520, 5506.

5) Hiroyuki Hirashita & **I-Da Chiang**, "Analytic models of dust temperature in high-redshift galaxies", 2022, MNRAS, 516, 1612.

4) **I-Da Chiang** et al., "Resolving the Dust-to-Metals Ratio and CO-to-H_2 Conversion Factor in the Nearby Universe", 2021, ApJ, 907, 29.

3) Eric W. Koch, **I-Da Chiang** et al., "Spatial power spectra of dust across the Local Group: No constraint on disc scale height", 2020, MNRAS, 492, 2663.

2) Dyas Utomo, **I-Da Chiang** et al., "The Resolved Distributions of Dust Mass and Temperature in Local Group Galaxies", 2019, ApJ, 874, 141.

1) **I-Da Chiang** et al., "The Spatially Resolved Dust-to-metals Ratio in M101", 2018, ApJ, 865, 117.

A full list of my publications is available at ORCID:0000-0003-2551-7148.

OBSERVING TIME AWARDED AS P.I.

VLA (2022B), "Connecting Gas and Dust: Mapping HI in 7 Herschel Galaxies", 28 hours *2022*

CERTIFICATIONS

Machine Learning Specialization *2023*
Coursera #2TT9G93HJ3LL (Offered by DeepLearning.AI & Stanford University)

Data Structures and Algorithms Specialization *2019*
Coursera #NKUTAK2CYZE6 (Offered by UC San Diego)

RESEARCH PRESENTATIONS

Contributed talk, "Resolved Maps of the CO-to-H_2 Conversion Factor in 41 Nearby Galaxies", East Asian Young Astronomers Meeting, Chiang Mai, Thailand *2024*

Contributed talk, "Tracing the kpc-scale CO-to-H_2 Conversion Factor with Dust in Galaxy Center", Illuminating the Dusty Universe: A Tribute to the Work of Bruce Draine, Florence, Italy *2023*

Contributed talk, "Kpc-scale properties of dust temperature in terms of dust mass and star formation activity", The 13th meeting on Cosmic Dust, Kitakyushu, Japan *2023*

Invited talk, "Quantifying the decrease of CO-to-H_2 conversion factor in galaxy centers", Taiwanese Theoretical Astrophysics Workshop II, Taipei, Taiwan *2022*

Colloquium, "Multiwavelength observations of dust, gas, and metals in the $z \sim 0$ universe", NCU, Taoyuan, Taiwan *2022*

Contributed talk, "Dust, gas, and metals: Observing Dust Evolution in Nearby Galaxies", Galaxy Evolution Workshop 2021, Tokyo, Japan *2021*

Colloquium, "Observations of Spatially Resolved Dust Evolution in Nearby Galaxies", ASIAA, Taipei, Taiwan *2021*

Contributed talk, "Dust, gas and metals: Resolving the Dust Life Cycle in the Nearby Universe", The AAS 235th Meeting, Honolulu, USA *2020*

Seminar, "Dust-to-Metals Relation in Nearby Galaxies", ASIAA, Taipei, Taiwan *2019*

Contributed talk, "The Variation of the Dust-to-Metals Ratio in Resolved Nearby Galaxies", Dusting the Universe, Tucson, USA *2019*

SERVICES

Colloquium & Lunch talk committee @ ASIAA *2022 - 2024*

Journal club "Galread" organizer @ ASIAA *2022 - 2024*

Postdoc representatives @ ASIAA *2021 - 2024*

OUTREACH

Cross-field seminar @ Institute of NanoEngineering and MicroSystems, NTHU – speaker *2024*

Sharing experience abroad @ Taipei Astronomy Workshop – panelist *2024*

Student seminar @ ASIAA – lecturer *2021*

Research in physics workshop for community college students @ UCSD – lecturer *2021*

Python workshop for physics undergrads @ UCSD – presenter *2019*

Life as a scientist @ Jianguo High School – speaker *2019*

STEM in Your Backyard: City Heights @ San Diego, USA – presenter *2018*

Tech Trek @ UCSD – presenter *2017*